

The Shim and the Proof

On Pirsig's Quality, Mathematics, and the Machine That Has No Attic

LeeMcCulloch-James

"We were both looking at the same thing, seeing the same thing, talking about the same thing, thinking about the same thing, except he was looking, seeing, talking and thinking from a completely different dimension."

Robert M. Pirsig, *Zen and the Art of Motorcycle Maintenance* (1974)

The Sham that is the Shim

The handlebars on John Sutherland's motorcycle have begun to slip. The bike is a BMW, expensive and beautifully made, and Pirsig's narrator who consider machines the way pro-mathematicians consider as scripture that proof-perfect, sees the repair at once. A shim: a thin strip of metal packed into the joint to take up the slack. The ideal material is sitting on the workbench in the form of an empty aluminium beer can.

A sham is a counterfeit and a shimmy is a feint, a shoulder thrown one way so the body may go the other; the shim is neither, and yet it borrows from both, for it gets you past the hump precisely by looking, for a moment, like the wrong thing to have done.

The metallurgy is worth dwelling on, because the narrator's whole case rests on it. Beer-can aluminium is soft and faintly tacky as metals go, which is exactly the property you want where two surfaces must grip. It resists corrosion: aluminium always carries a thin skin of oxide that seals the surface and prevents any further oxidation the instant it is exposed. Cut a strip, trim it, seat it, and the joint is fixed. A German master mechanic with fifty years behind him would look at the result and call it apt.

He offers the shim to John who recoils from it. For John has never seen that metallurgic state for anything but what it first appears. He sees a beer-can going into a precision machine, and refuses to accept that as anything but an unhappy marriage. The

notion generates a dull thud in him, while the narrator is left perplexed with their lack of accordance on its obvious *affordance*.

Two Dimensions

Pirsig's narrator was seeing what the shim *means*: its function, its fitness to resolve the underlying form of the problem. John was seeing what the shim *is*: a discarded form fit for another function, an insult to his highly machined machine.

Pirsig names the narrator's mode classical: the world apprehended through form, through analysis, through the patient extraction of cause and effect. John's mode as the appreciator is romantic the world taken in at its surface, intuitively, all at once.

The easy move is to take sides. Call John a sentimentalist who would sooner ride a broken bike than fit an ugly repair; call the narrator a cold reductionist who has analysed the soul out of the function. Pirsig wants rather to pitch the classical and the romantic as two halves of a single act of attendance.

What is dark about John is that he cares about technology a great deal. The romantic mode is the only one available to him, and when it fails, when the machine will not simply *groove* for the drummer in him he has nowhere to ride. His sensitivity, cut off from analysis, leaves him a spectator of the machinery his life runs on: alert to it, but strangely estranged from it.

Where Quality Lives

Here is the claim Pirsig's book is built to support. *Quality* is not a property of the shim and not a feeling inside John or the narrator. Rather it lives in the relationship between a person and the work in front of them. That is, in the quality of the attention they bring to bear. The same strip of aluminium can be a beautiful customized repair or an ugly one borne of serendipity. The machine is agnostic to its provenance.¹

This is why Pirsig can say, that the Buddha sits as comfortably in the gears of a transmission as in the petals of a flower. The gears deserve the same care and attention the flower gets. Quality asks one question of a functioning form, and it is always the same question: are you purposefully here.

¹Here being from the hands of the classical seeing-into-form rather than the romantic seeing-of-surface.

The Maths Was Always a Shim

If we move our attendance to mathematics, it is worth asking what of such matters shim now resonates more than ever. There is a comfortable picture, often espoused by the likes of Roger Penrose, no less in which mathematics is a fixed Platonic territory, eternal and complete, with the mathematician as an explorer charting rooms that were always there, so that the history of the subject becomes a record of which rooms got visited and in what order. As a handy concomitant reality instantiates from that which it conjured. It is a tidy if circular picture.

We arrived at the number notion pragmatically, likely first through counting days (before sheep presumably). Then we counted the gaps within days, until the gaps turned out to have gaps of their own and the rationals became unavoidable; and when the diagonal of a unit square slipped through our Pythagorean rational notions, something Real had to be built that could catch such surds. Each widening of the number concept came from a real problem about real magnitudes, noticed by creatures with decimally primed hands: that sequence of abstractions indelibly carries our fingerprints.

None of this makes subsequent more sophisticated structures any more invented. The [Eisenstein integers](#), $\mathbb{Z}[\omega]$, are indifferent to whether anyone ever wrote them down, and their arithmetic is as fixed as anything in Plato's heaven, so that the open question is about us, our relationship to them, rather than about them. Which of the manifold of real structures we developed, and which problems felt urgent enough to name and pursue, is a product of the human situation and about what a hundred thousand years of embodied life made salient. The territory may be mostly ephemeral and unrealisable² while the path through and to it was most certainly cut by our real bodies. It is this that we have learned to reside within that we might call *physically curated Platonism*.

There is a particular move in mathematics that earns the name *shim* more than any metaphor has a right to, which is the moment a proof reaches outside its own domain, borrows a structure milled in some quite different corner of reality, packs it into the gap, and gets past a hindrance the native tools could not clear. One of its cleanest instances while being newsworthy of late is perhaps Furstenberg's proof of Szemerédi's theorem. The theorem itself is a statement about the integers, asserting that any set of whole numbers thick enough to occupy a positive fraction of the number line must contain arithmetic progressions of every finite length, runs of evenly spaced numbers as long as you

²Hardy, (not the Thomas variety) gave us an early warning here, having prized number theory precisely for its uselessness, the purest of the pure and untouched by application, only for cryptography within his own century to make that useless beauty into the load-bearing mathematics of global commerce. He had arrived late enough in a long history of motivated attention that he could forget the origins entirely and mistake the forgetting for purity, while RSA sat waiting in the abstraction the whole time.

please. Szemerédi had already proved it by a combinatorial argument of famous brutality, a labyrinth of cases that convinces without illuminating.³ Furstenberg then reached onto an entirely different shelf and brought back a tool from *ergodic theory*, which is the branch of mathematics that studies dynamical systems over the long run, systems that evolve by some repeated transformation while preserving a notion of volume or probability, asking what happens to a region as the process is iterated again and again toward infinity. The governing fact of that field is a recurrence principle, the result that in such a system almost every state must eventually return arbitrarily close to where it began, and return not once but infinitely often. Furstenberg showed that the combinatorial claim about progressions in the integers is, when properly translated, the very same statement as a claim about such a system returning near its starting configuration under repeated shifts, so that the recurrence becomes the packing strip that takes up the slack. Nothing in a question about evenly spaced integers gives any warning that flows, invariant measures and the long-run behaviour of dynamical systems should have anything to say, and yet the borrowed piece seats into the joint as though it had been cut for it.

The Complex number extension to the field of the Reals which we reached above, rather inverts our metaphor, though in a way worth pausing on. They would be the ideal borrowed shim if physics had needed them first and mathematics had merely reached across to take them, but the chronology runs backwards, since they were forced on Cardano and Bombelli in the sixteenth century by nothing more exotic than the algebra of cubic equations, where a perfectly real solution can sometimes be reached only by passing through the square root of a negative quantity that cancels itself before the end. They were a convenience, a packing strip cut to clear a hump in algebra, three full centuries before Maxwell and then the founders of quantum mechanics discovered that electromagnetism and the amplitudes of the quantum world actually run on them natively. The borrow turned out to have been structural all along, which is the unsettling mirror image of the shim, the strip you cut for mere convenience revealing itself, much later, to have been load-bearing in the fabric of things. The prime number theorem carries the same doubleness in miniature, since the statement is purely about how the primes thin out

³The same hump has since been cleared by more than one borrowed strip. Timothy Gowers, in the work that won him the Fields Medal in 1998, re-proved the theorem from a wholly different direction, inventing what is now called higher-order Fourier analysis and obtaining along the way the first usable quantitative bounds, where Szemerédi’s combinatorial proof had yielded almost none [1, 2]. The Gowers norms introduced there have become central machinery for the whole field of additive combinatorics, and the area remains strikingly active: the quantitative bounds for the three-term case have tumbled in recent years through the breakthrough of Kelley and Meka [3] and its sharpening by Bloom and Sisask [4], and one of the field’s long-standing conjectural pillars, the Polynomial Freiman–Ruzsa conjecture of Marton, was settled in characteristic two by Gowers, Green, Manners and Tao [5], in a proof notable for having been formally verified in the LLM driven Lean proof assistant. That a single theorem should fall to native combinatorics, to an ergodic borrow, and to a Fourier-analytic one is itself the shim point in miniature.

among the integers and was first proved by pushing the Riemann zeta function out into the complex plane and reasoning about where it refuses to vanish, a contour in a borrowed plane doing work the primes never asked for, until Erdős and Selberg found an elementary proof that removed the complex machinery entirely. That later proof has something of Lee Sedol’s answering move about it, the demonstration that a human path through the same position existed after all, harder to find and stranger in its way, proving that the alien route through the complex plane had been convenient rather than necessary, with both proofs left sitting on the same board.

The love of an elegant proof may therefore be closer to John’s flinch than mathematicians care to admit. They may all recoil from an ugly proof, whether a brute case-analysis or a thousand-page machine verification or an argument that grinds where it ought to sing, before they have properly checked it, reacting romantically with something very like John’s dull thud. The professional mathematician’s flinch has at least been calibrated by long contact with the territory, lest they quietly forget the contact that sent even a Ramanujan down a wrong path. The romantic and the classical run together in the working mathematician, and in the best of them it would appear any a divisive seam has long since healed.

The LLM Machine that clears the Attic

There is a new machine on the Mathematician’s workbench now, and it changes the terms of everything that has come before. Andrew Wiles spent seven years in an attic, in something close to secrecy, pressing his mind against Fermat’s Last Theorem until it gave, and what the community received when he finally came down included the attic itself, since the seven years were part of the meaning, a human story about one mind meeting one problem and not breaking under it. We learned that the theorem was true, and we learned at the same time what it had cost.

A large language model now resolves combinatorial problems that Erdős posed eighty years ago, problems that defeated people who gave them their lives, and the proofs check out. What unsettles is not the possibility of error but the absence of an attic, since there was no struggle, no secrecy, and no biography of attention behind the result. A system has absorbed the accumulated selections of ten thousand mathematical lives and located, in the geometry of that absorbed space, paths that no embodied mind had ever walked, so that the result arrives with little of our selves attached to it at all.

This does not make the result worth less, though one may notice that we want it to. What it changes is what we are *for* and when we *do* mathematics, in much the way that the parlour games we set aside as a measure of our cleverness have changed under the

same new attention. The chess case came first, and it was the gentler of the two. When [Deep Blue](#) beat Kasparov over six games in 1997, becoming the first machine to defeat a reigning world champion under tournament conditions, it did so by a margin of three and a half to two and a half, and it did so essentially by force, evaluating something like two hundred million positions a second. The intelligence on display was engineered search, opening preparation and sheer hardware, so that what the defeat told us was bounded and almost reassuring, namely that machines can out-calculate us even in one of the most cerebral of our games. Chess survived the verdict and its meaning merely shifted, becoming a human performance under a known ceiling, watched the way we watch a pianist play Beethoven in full knowledge that the definitive recording already exists.

The Go case, twenty years later, of course cuts deeper, because it touches the faculty we had assumed was ours alone. When [AlphaGo](#) beat Lee Sedol four games to one in Seoul in 2016, what mattered was less the score than Move 37 of the second game, a stone placed on the fifth line that DeepMind's own network reckoned a human professional would play perhaps once in ten thousand times. The commentators assumed it was a mistake, and even the engineers hesitated, and then over the following hours it resolved into something that looked less like an error than a profundity, an idea arriving from outside the several thousand years of accumulated human Go wisdom rather than from within it. The delightful Lee, who had begun the match confident in the irreducibility of human intuition, said afterwards that he had thought the machine merely calculated probabilities until that move changed his mind and convinced him the thing was creative. The asymmetry with chess is the whole point, since Deep Blue defeated our calculation while AlphaGo seemed to defeat our intuition, producing strategy that looked non-human and yet meant something, which is exactly the shape of the worry about a proof that arrives with no attic behind it.

What rescues the Go story from despair is Game Four, where Lee answered with a move of his own, the wedge at Move 78 since remembered as God's Touch, a placement just as improbable as AlphaGo's had been, reckoned afterwards at the same one in ten thousand, and it was the single game he won. The lesson I take from the pairing is not that the machine reached a ceiling we can be comforted by, but that a quality human had the capacity for the genuinely inventive move when confronted by a system foe that could also make one; those two improbabilities sat on the same board across two games.

Pirsig told us where quality would relocate, half a century earlier. If Quality was always in the relationship of attention, then the machine that produces the theorem has taken nothing from us that was ever ours to begin with. On the far side of this, mathematics may become what motorcycle maintenance already was for the narrator, a practice taken up because the attention it demands is the point, with or without results, something close

to monastic, and the pure mathematician and the Zen mechanic may find themselves working the same bench.

This consolation may be merely convenient, since a discipline whose novelty has been outrun would naturally tell itself that the novelty was never what mattered. The monastic reading might be a genuine wisdom, or it might be just the most sophisticated available form of a blanket curse.

Epistemic Atrophy

There is a word now for the thing underneath the worry, what Lucy Gill-Simmen has called *epistemic atrophy* [6], and the faculty it points to is the very one that Quality was supposed to require, namely the capacity to hold attention steady on the cusp between what we know, what we do not yet know, and what we may never know. Gill-Simmen describes the danger as the slow erosion of our willingness to do the harder work of understanding what is real, the instinct to distrust the fluent answer, and the patience to sit with the discomfort of not-yet-knowing, a phrase I borrow from her almost intact because I cannot improve on it, since it is precisely that patience which the present argument turns upon. The claim is that habitual reliance on these machines wastes the capacity in the way a limb wastes inside a cast, and the cusp it concerns has three regions that the machine touches quite differently.

The known is the oldest bargain and the least alarming, since we have been offloading settled fact to external stores ever since writing, and Plato was already uneasy about it in the *Phaedrus*⁴, convinced that letters would hollow out memory. Memory did change, and we are still here, so that outsourcing the known is something we have survived and flourished because of before, and there is no obvious reason to think this round different in kind.

The unknown is where it begins to bite, because sitting inside an open problem without an answer, tolerating the confusion long enough for some structure to surface, looks like a trainable capacity, a muscle that grows under load. Wiles needed the seven years partly because the sitting was itself the work, and the mind that could finally see the proof was built by the long years of not seeing it, so that a machine which hands you a plausible path before you have sat there long enough may give you the theorem while quietly starving

⁴The title carries a second charge here that is too apt to pass over. Pirsig gave the name *Phaedrus* to his narrator's former self, the fierce pre-treatment personality who pursued Quality to the edge of madness and was, on the narrator's account, erased by court-ordered electroconvulsive therapy, so that the man writing the book is shadowed throughout by the thinker he used to be. That an argument about the externalising of memory should come to rest on the *Phaedrus* is therefore a double exposure, Plato's dialogue about memory given over to the written mark laid over Pirsig's *Phaedrus* who is himself a memory deliberately wiped, the very erosion the section fears enacted once in ink and once in a man.

the faculty that would have grown in the waiting.

The potentially unknowable is the region an LLM is least equipped to speak about honestly. LLMs are built to be useful, which is to say built to offer another angle, another reframing, another move, and that disposition runs directly against the unknowable as a category, because to dwell at a genuine limit you have to recognise it as a limit and remain there without resolving it, in the apophatic patience the mystics cultivated, the *via negativa*, the willingness to let a horizon stay a horizon. A thing that always has a next suggestion trains you out of standing still at the edge.

And yet we do not want to hand the worry an easy victory, because there is a real case to be made on the other side. Every cognitive prosthesis we have ever adopted drew this same complaint, and the complaint was never simply wrong, since memory did weaken and mental arithmetic did fade, and yet the freed attention did not vanish so much as move elsewhere. The calculator did not end mathematics, it cleared the desk so that harder questions could be asked, and a generation that never learned long division by hand has produced people who sit at genuine frontiers, so that atrophy in one register has before now been the precondition for development in another. It is at least possible that offloading the known and the merely difficult is what frees a mind to spend its scarce attention at the unknowable edge, rather than what keeps it away, and that the same machine which shortens the sitting also clears the ground around it.

The optimistic case is real, and could be held sincerely but the worry is sharper than the optimism precisely because its losses are the kind that hide, since a frontier you slowly stop walking toward leaves no failed output behind to flag what has gone. Both may be true at once, the tool that extends the reach of attention and the habit that wastes the muscle.

Who Is John Now

Persig's John is still very much with us, and he is often the cleverest person in the room. He will feel, when a model's output is hollow or alive, having a genuine romantic sensitivity to the thing, and yet he relates to it only as an instrument to groove with or to curse. John would phrase the shim of the machine as *just autocomplete*, the beer-can shim reaction in modern dress, true enough on the face of it.

The narrator's reply would be the one he gave about the bike. You need not become a mechanic, and you need not implement a transformer or derive the attention mechanism by hand, but you do need to know enough of what sits under the bonnet to stop being mystified by failures that have perfectly plain explanations. The model holds no continu-

ous idea of a word, so that *unbelievable* and *un·believe·able* can be quite different objects to it, and where you place a constraint in the context window matters mechanically, and the thing performs a weighted nearest-neighbour completion across an enormous space rather than retrieving truth from a vault. These are the beer-can-shim facts, ungla-morous and slightly deflating, and they are the only thing that turns a romantic flinch into productive attention.

The person who manages to hold both at once, grooving with the output while asking what produced it, is standing exactly where Pirsig pointed, with the classical and the romantic held together and neither allowed to drive the other out. That is where Quality lived in the gears of the transmission, and it is where Quality lives now in the collaboration between a human attention that can be present and a machine that, for all its fluency, cannot.

The shaping of mathematics continues in any case, curated as ever and cut by need, though now accelerated past anything that bodies could manage on their own, since we have built something that ranges across our accumulated selections with no body, no friction, and no hour in the attic. The question it hands back to us is the one the Perig’s narrator was holding in front of a slipping handlebar with a beer can in his hand, which is what kind of attention we mean to practise, and that is a question the machine will not answer for us. It may be the last part of the work that stays ours, though I am not entirely sure even of that.

References

- [1] W. T. Gowers, *A new proof of Szemerédi’s theorem for arithmetic progressions of length four*, Geometric and Functional Analysis ([GAFA](#)) **8** (1998), no. 3, 529–551.
- [2] W. T. Gowers, *A new proof of Szemerédi’s theorem*, Geometric and Functional Analysis ([GAFA](#)) **11** (2001), no. 3, 465–588. doi:10.1007/s00039-001-0332-9.
- [3] Z. Kelley and R. Meka, *Strong bounds for 3-progressions*, preprint, 2023. [arXiv:2302.05537](#).
- [4] T. F. Bloom and O. Sisask, *An improvement to the Kelley–Meka bounds on three-term arithmetic progressions*, preprint, 2023. [arXiv:2309.02353](#).
- [5] W. T. Gowers, B. Green, F. Manners and T. Tao, *On a conjecture of Marton*, Annals of Mathematics (2) **201** (2025), no. 2, 633–782. [arXiv:2311.05762](#).
- [6] L. Gill-Simmen, *AI doesn’t just help us think, it thinks instead of us: what this means for the process of learning*, The Conversation (UK), June 2026.